

Statistical Analysis Explanation for a Survey of Job Satisfaction Among Anaesthesia Registrars and Consultants working in Academic Hospitals in Gauteng Province, South Africa

Participant characteristics

Initial data cleaning involved renaming variables for clarity, recoding categorical responses into ordered factors, and collapsing certain categories to ensure analytical feasibility (e.g., merging “Supernumerary Registrar” into “Registrar” and grouping age ranges into broader categories). Missing values were handled conservatively by retaining cases with sufficient information for the relevant analysis, and all transformations were performed using reproducible, script-based workflows

The study population comprised 161 anaesthesia registrars and consultants working at academic hospitals in Gauteng province, South Africa (Table 1). Of those who completed the survey, registrars constituted most of the respondents ($n = 116$, 81%), with consultants comprising the remaining participants ($n = 28$, 19%).

In terms of duration in their current role, a notable proportion (38%) reported more than three years of tenure, while approximately one-quarter (24%) had between two- and three-years’ experience. Participants newer to their positions, those with one to two years and those between six months to one year, made up 21% and 17% of the sample, respectively.

Most registrars indicated they were in the earlier stages of their specialist training, with 78% having completed the Fellowship of the College of Anaesthetists Part 1 (FCA Part 1) and 15% having achieved FCA Part 2. A smaller segment (5.6%) reported other qualifications, and only 1.4% held a Diploma in Anaesthesia (DA).

The gender distribution indicated a predominantly female cohort (67%), while male respondents accounted for 33%. One participant (0.7%) chose not to disclose their gender. Age-wise, more than half (58%) of the respondents fell within the 25–35-year age range, suggesting a younger demographic overall. Smaller proportions belonged to the 36–40-year (19%) and the 41+ year groups (24%).

Most respondents (67%) were married, and approximately one-quarter (27%) were single. A smaller group (6.5%) reported living with a partner. Additionally, nearly two-thirds (63%) of respondents indicated they were parents. Family proximity varied significantly, with 44% having immediate family living in different households but within the same town, and 40% in a different province. Only a small minority (1.6%) reported having immediate family living internationally.

Table 1. Participant characteristics

Variable	N = 161 [†]
Position	
Consultant	28 (19%)
Registrar	116 (81%)
Time in post	
Less than 6 months	0 (0%)
6 months to 1 year	25 (17%)
1 to 2 years	30 (21%)
2 to 3 years	34 (24%)
More than 3 years	55 (38%)
Qualification	
MBBCh/MBChB/MBBS	0 (0%)
DA	2 (1.4%)
FCA Part 1	112 (78%)
FCA Part 2	22 (15%)
Other	8 (5.6%)
Gender	
Male	47 (33%)
Female	96 (67%)
Prefer not to disclose	1 (0.7%)
Age range	
25–35	83 (58%)
36–40	27 (19%)
41+	34 (24%)
Marital status	

Variable	N = 161 ¹
Married	92 (67%)
Single	37 (27%)
Widowed	0 (0%)
Divorced	0 (0%)
Living with a partner	9 (6.5%)
Parent	90 (63%)
Family proximity	
Within same household	0 (0%)
Different households, same town	27 (44%)
Different towns, same province	9 (15%)
Different province	25 (40%)
Different country	1 (1.6%)
¹ n (%)	

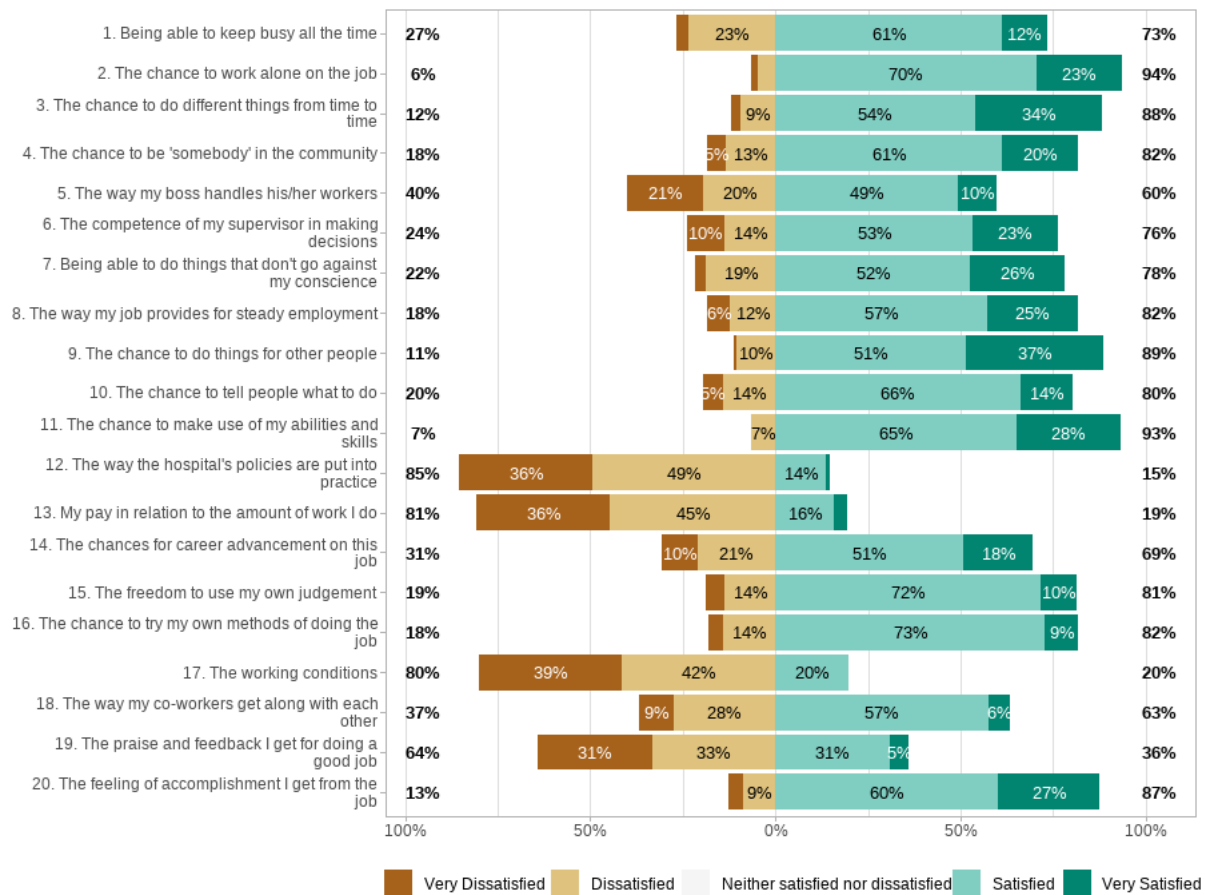
MSQ Item Responses

The Likert plot (Figure 1) visually summarises participants' responses to individual items from the Minnesota Satisfaction Questionnaire (MSQ), illustrating areas of satisfaction and dissatisfaction among anaesthesia registrars and consultants. This figure was generated by first converting the MSQ responses into a numeric, ordered Likert-scale format. Each item was subsequently summarised by calculating the proportion of responses within each satisfaction category (ranging from "Very Dissatisfied" to "Very Satisfied"). Finally, these results were visualised using a diverging stacked-bar plot.

The figure highlights strong satisfaction with job autonomy and skill utilisation, notably for items such as *"The chance to work alone on the job"* (94% satisfied or very satisfied) and *"The chance to make use of my abilities and skills"* (93% satisfied or very satisfied). Conversely, significant dissatisfaction emerges around institutional factors, particularly *"The way the hospital's policies are put into practice"* (85% dissatisfied or very dissatisfied), *"My pay in relation to the amount of work I do"* (81% dissatisfied or very dissatisfied), and *"The working conditions"* (80% dissatisfied or very dissatisfied). The item addressing

feedback, *"The praise and feedback I get for doing a good job"*, also demonstrates substantial dissatisfaction (64%), underscoring an actionable area for management intervention

Figure 1. Distribution of satisfaction responses for the 20 MSQ items



Responses from the Minnesota Satisfaction Questionnaire (MSQ) were initially recoded into a numeric five-point Likert scale format ranging from "Very Dissatisfied" to "Very Satisfied". To present these data clearly, responses were reshaped from a wide format where each survey item was a separate column to a long format, capturing each respondent's answer as an individual record. Subsequently, these responses were cross tabulated to produce a summary table, presenting the percentage distribution of responses for each MSQ item clearly by satisfaction level. The resulting table provides a concise overview of how respondents evaluated each facet of their job satisfaction.

The summary table (Table 2) shows clear variation in satisfaction across different job facets. Most respondents expressed satisfaction with their ability to use their skills and maintain

busy work schedules. Items such as *"The chance to make use of my abilities and skills"* and *"The freedom to use my own judgement"* received predominantly positive ratings (satisfied or very satisfied exceeding 70%). Conversely, institutional policies, compensation, and physical working conditions elicited substantial dissatisfaction, with over 80% of respondents rating these aspects negatively (either dissatisfied or very dissatisfied).

Table 2. Percentage distribution of responses for each MSQ item

	Response					Overall
	Very Dissatisfied	Dissatisfied	Neither satisfied nor dissatisfied	Satisfied	Very Satisfied	
Question						
1. Being able to keep busy all the time	3 (3.3%)	21 (23%)	0 (0%)	55 (61%)	11 (12%)	90 (100%)
2. The chance to work alone on the job	2 (1.9%)	5 (4.6%)	0 (0%)	76 (70%)	25 (23%)	108 (100%)
3. The chance to do different things from time to time	3 (2.6%)	11 (9.4%)	0 (0%)	63 (54%)	40 (34%)	117 (100%)
4. The chance to be 'somebody' in the community	5 (5.1%)	13 (13%)	0 (0%)	60 (61%)	20 (20%)	98 (100%)
5. The way my boss handles his/her workers	18 (21%)	17 (20%)	0 (0%)	43 (49%)	9 (10%)	87 (100%)
6. The competence of my supervisor in making decisions	11 (10%)	15 (14%)	0 (0%)	58 (53%)	25 (23%)	109 (100%)
7. Being able to do things that don't go against my conscience	3 (3.0%)	19 (19%)	0 (0%)	53 (52%)	26 (26%)	101 (100%)
8. The way my job provides for steady employment	7 (6.1%)	14 (12%)	0 (0%)	65 (57%)	28 (25%)	114 (100%)
9. The chance to do things for other people	1 (0.9%)	12 (10%)	0 (0%)	59 (51%)	43 (37%)	115 (100%)

	Response					Overall
	Very Dissatisfied	Dissatisfied	Neither satisfied nor dissatisfied	Satisfied	Very Satisfied	
10. The chance to tell people what to do	3 (5.4%)	8 (14%)	0 (0%)	37 (66%)	8 (14%)	56 (100%)
11. The chance to make use of my abilities and skills	0 (0%)	8 (6.7%)	0 (0%)	78 (65%)	34 (28%)	120 (100%)
12. The way the hospital's policies are put into practice	34 (36%)	47 (49%)	0 (0%)	13 (14%)	1 (1.1%)	95 (100%)
13. My pay in relation to the amount of work I do	41 (36%)	51 (45%)	0 (0%)	18 (16%)	4 (3.5%)	114 (100%)
15. The freedom to use my own judgement	5 (4.9%)	14 (14%)	0 (0%)	73 (72%)	10 (9.8%)	102 (100%)
16. The chance to try my own methods of doing the job	4 (4.0%)	14 (14%)	0 (0%)	72 (73%)	9 (9.1%)	99 (100%)
17. The working conditions	37 (39%)	40 (42%)	0 (0%)	19 (20%)	0 (0%)	96 (100%)
19. The praise and feedback I get for doing a good job	24 (31%)	26 (33%)	0 (0%)	24 (31%)	4 (5.1%)	78 (100%)
20. The feeling of accomplishment I get from the job	4 (3.9%)	9 (8.7%)	0 (0%)	62 (60%)	28 (27%)	103 (100%)
Total	205 (11%)	344 (19%)	0 (0%)	928 (51%)	325 (18%)	1,802 (100%)

Minnesota Satisfaction Questionnaire Scale Analysis

Cronbach's α quantifies the degree to which items within a scale measure the same underlying construct; values above 0.70 are generally regarded as acceptable for research

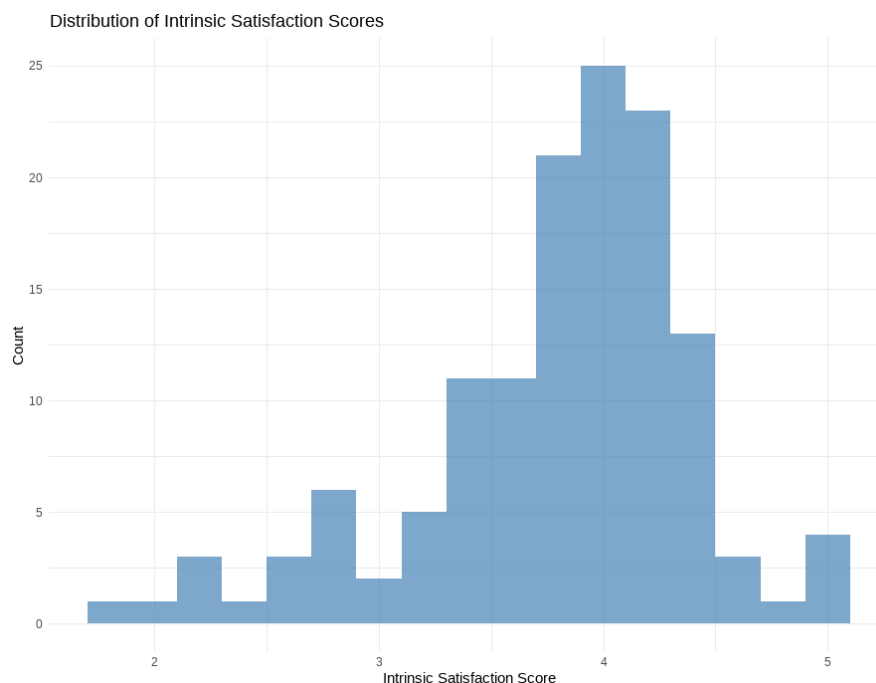
purposes, while values above 0.80 are commonly interpreted as “good” and those approaching or exceeding 0.90 as “excellent.”

All three scales are sufficiently reliable for group-level comparisons and subsequent multivariable modelling.

Scale	No. items	Cronbach's α (95 % CI)
Intrinsic	12	0.84 (0.81 – 0.88)
Extrinsic	8	0.81 (0.77 – 0.86)
General	20	0.89 (0.87 – 0.92)

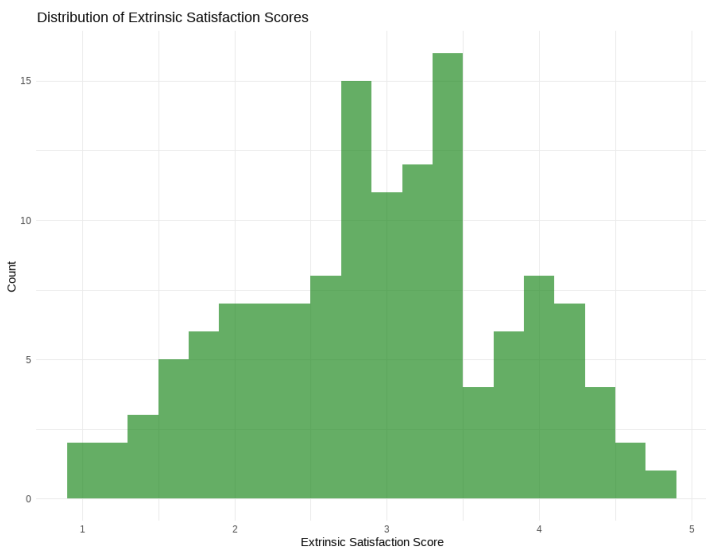
The histogram of intrinsic satisfaction scores is tightly clustered between 3.5 and 4.5, peaking just above 4.0. Only a handful of observations fall below 3.0 and virtually none exceed the theoretical maximum of 5.0. This left-skewed concentration indicates high and relatively homogeneous intrinsic satisfaction, that is, most respondents feel positively about job features they can directly control (e.g., autonomy, variety, use of skills). The limited tail below 3.0 suggests only a few individuals experience marked intrinsic dissatisfaction.

Figure 2A – Intrinsic Satisfaction



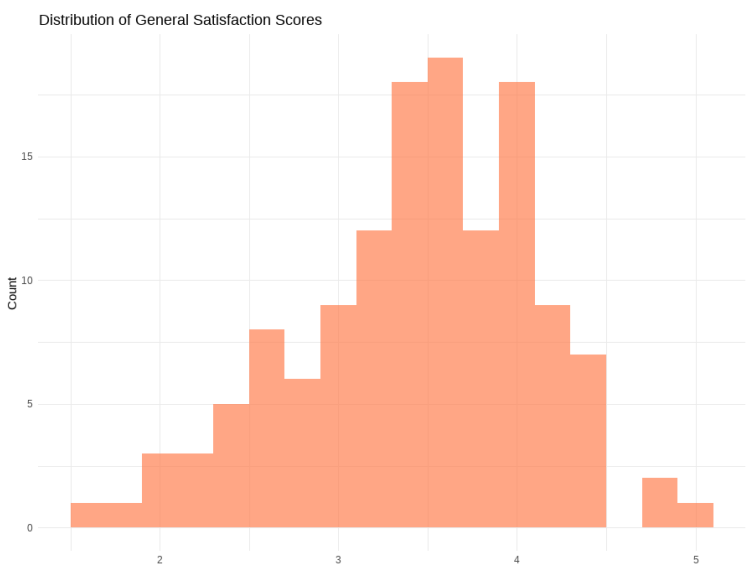
Extrinsic scores span a much broader range, from approximately 1.0 to just above 4.0, with a primary mode around 3.0 and a noticeable shoulder near 2.0. The distribution is flatter and right skewed, reflecting greater heterogeneity and generally lower endorsement of extrinsic aspects such as pay, supervisory support, working conditions, and organisational policy. In practical terms, respondents' views on these externally determined factors vary widely, from extreme dissatisfaction to moderate satisfaction, underscoring systemic issues rather than individual preferences.

Figure 2B. Extrinsic Satisfaction



The overall (general) score combines intrinsic and extrinsic items, producing a near-normal bell shape centred around 3.5–3.8. This composite is moderately high on average, but its wider spread compared with the intrinsic scale reflects the influence of lower extrinsic ratings. The right tail extends toward 5.0, yet the left tail dips as low as roughly 2.0, signalling that while most participants report reasonable overall satisfaction, a non-trivial minority remain discontented.

Figure 2C. General Satisfaction

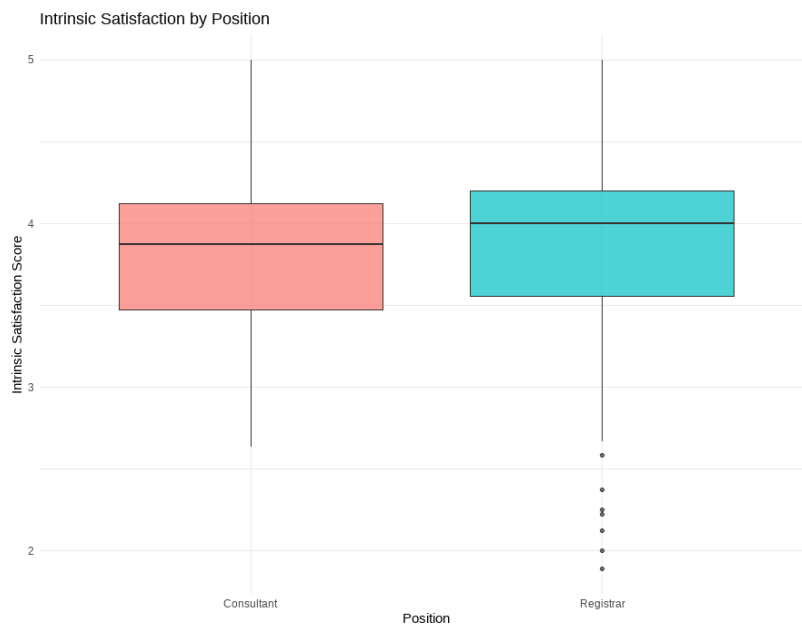


Boxplots

Each boxplot compares **consultants** and **registrars** on the three Minnesota Satisfaction Questionnaire (MSQ) composites. The box denotes the inter-quartile range (IQR; 25th–75th percentiles) and the bold line marks the median. Whiskers extend to the most extreme values that lie within $1.5 \times \text{IQR}$ of the box; points beyond these limits are plotted individually and represent outliers, unusually high or low scores relative to the bulk of the data.

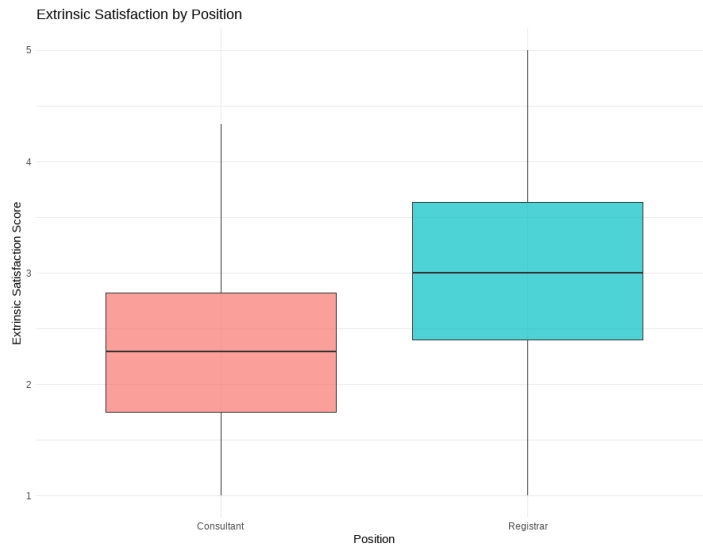
For intrinsic satisfaction, medians are high in both groups (≈ 4.0), and IQRs largely overlap, indicating uniformly strong intrinsic motivation. A handful of low-scoring registrars (outliers ≤ 2.3) suggest isolated cases of intrinsic disengagement.

Figure 3A. Intrinsic satisfaction



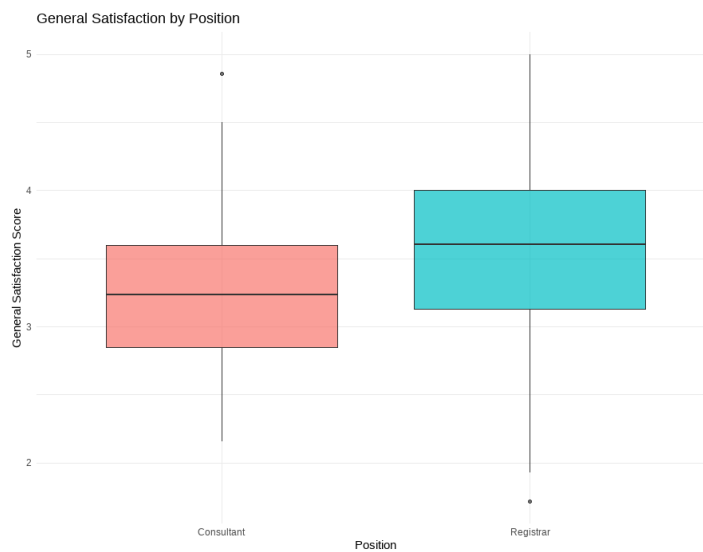
Registrars sit ~ 0.7 units higher in median than consultants (3.0 vs 2.3) and display a broader upper whisker, implying that trainees experience better extrinsic support than senior staff. Consultants show several low outliers near 1.0, underscoring pronounced dissatisfaction with organisational factors in a subset of specialists.

Figure 3B. Extrinsic satisfaction



The global pattern mirrors the extrinsic results: registrars' median (≈ 3.5) exceeds consultants' (≈ 3.3). Extreme values at both tails (e.g., a consultant near 4.8 and a registrar near 2.1) are flagged as outliers, but the central spread again suggests that differences in extrinsic conditions drive the modest gap in overall satisfaction.

Figure 3C. General satisfaction



Composite Satisfaction Scores by Clinical Position

Mean intrinsic scores are virtually identical between consultants (3.81) and registrars (3.79). The negligible mean difference (+0.01) is accompanied by a wide confidence interval that spans zero (-0.25 to +0.28) and a non-significant p-value (> 0.90), indicating no evidence of a positional difference in how respondents value autonomy, skill use, or other intrinsic facets of their work.

Registrars report a substantially higher mean extrinsic score (3.07) than consultants (2.50). The estimated mean gap of -0.57 points (95 % CI: -0.91 to -0.22) is statistically significant ($p = 0.002$). Consultants are markedly less satisfied with pay, supervisory support, working conditions, and organisational policies than their trainee counterparts. Because the CI does not include zero and the effect size approaches two-thirds of an SD, this is both statistically and practically meaningful.

When intrinsic and extrinsic items are combined, registrars score on average 0.26 points higher than consultants (3.50 vs 3.24). Although the point estimate favours registrars, the 95 % CI (-0.54 to +0.02) narrowly crosses zero and the p-value (0.070) exceeds the conventional 0.05 threshold. The global satisfaction difference is suggestive but does not reach statistical significance, likely because high intrinsic satisfaction among consultants compensates for their lower extrinsic ratings.

Table 3. Summary statistics of Composite Satisfaction Scores by Clinical Position

Characteristic	Overall N = 144 ¹	Consultant N = 28 ¹	Registrar N = 116 ¹	Difference ²	95% CI ²	p-value ²
Intrinsic satisfaction score	3.80 (0.61)	3.81 (0.60)	3.79 (0.61)	0.01	-0.25, 0.28	>0.9
Extrinsic satisfaction score	2.96 (0.86)	2.50 (0.76)	3.07 (0.85)	-0.57	-0.91, -0.22	0.002
General satisfaction score	3.45 (0.66)	3.24 (0.63)	3.50 (0.66)	-0.26	-0.54, 0.02	0.070

Abbreviation: CI = Confidence Interval

¹ Mean (SD)

² Welch Two Sample t-test

Univariable Ordinal-Logistic Models Predicting Intention to Leave

The ordinal logistic regression analysis assessed how various characteristics (position, gender, age range, marital status) relate to clinicians' intention to leave their current post, categorised as an ordered scale ("No", "Thinking about it", "Yes"). Results are expressed as odds ratios (OR), which quantify the odds of being in a higher category of intention to leave, compared to a reference group. The proportional odds assumption was assessed and was not violated based off of a non-significant result of Brant's test.

Registrars had an OR of 0.23 (95% CI: 0.10–0.50) compared to consultants. This indicates that registrars had 77% lower odds of expressing stronger intentions to leave, which was statistically significant as the confidence interval did not cross 1. Consultants are therefore significantly more likely to consider leaving or actively seek new positions.

Female clinicians (OR = 0.81, CI: 0.39–1.72) and those preferring not to disclose their gender (OR = 2.42, CI: 0.09–68.8) did not differ significantly from males. The wide confidence interval for the "Prefer not to disclose" group suggests substantial uncertainty due to the small sample size in this category.

Clinicians aged 36–40 (OR = 2.34, CI: 0.94–5.75) and those older than 41 (OR = 2.22, CI: 0.98–5.00) demonstrated higher odds of intending to leave compared to younger colleagues (25–35 years), though these differences narrowly missed statistical significance.

Neither single clinicians (OR = 0.83, CI: 0.38–1.84) nor those living with a partner (OR = 1.51, CI: 0.38–6.05) significantly differed from married colleagues in their likelihood to consider leaving.

Table 4. Ordinal-Logistic Models Predicting **Intention to Leave**

Characteristic	OR	95% CI
Position		
Consultant	—	—
Registrar	0.23	0.10, 0.50

Characteristic	OR	95% CI
Gender		
Male	—	—
Female	0.81	0.39, 1.72
Prefer not to disclose	2.42	0.09, 68.8
Age Range		
25–35	—	—
36–40	2.34	0.94, 5.75
41+	2.22	0.98, 5.00
Marital Status		
Married	—	—
Single	0.83	0.38, 1.84
Living with a partner	1.51	0.38, 6.05
Abbreviations: CI = Confidence Interval, OR = Odds Ratio		